

Mohamed Maher Mahfouz El-Abyad

Senior Software Engineer

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PROFILE SUMMARY

Senior Software Engineer with 8 years of experience designing and scaling systems using PHP (Laravel). Led the adoption of Docker and Kubernetes, improving CI/CD pipelines and system reliability. Passionate about building AI-powered features and driving cross-functional innovation using Go, PHP, and modern cloud tools.

WORK EXPERIENCE

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|--|-----------------------------------|
| Senior Software Engineer
<i>Trend</i> | 07/2025 – Present
Cairo, Egypt |
| <ul style="list-style-type: none">• Own the backend of TrendX, a bilingual survey & community platform — a Laravel 11 REST API with 190+ endpoints backed by MySQL, Redis, Elasticsearch, and a ClickHouse analytics pipeline.• Introduced Docker containerization and Kubernetes orchestration for deployment pipelines, and built Kafka consumers and Redis-backed queues/caching to handle high-traffic feeds and event streams.• Maintain an automated PHPUnit test suite with CI-gated pull requests, and shipped LLM-powered features (Anthropic API) for survey question generation, translation, and content moderation. | |
| Senior Software Engineer
<i>YouMats</i> | 08/2024 – 06/2025
Dubai, UAE |
| <ul style="list-style-type: none">• Leading the development of a new, large-scale system using PHP, Laravel, and PostgreSQL, leveraging the full Laravel ecosystem for robust, scalable architecture. | |
| Senior Software Engineer
<i>Dimofinf</i> | 05/2024 – 08/2024
Cairo, Egypt |
| <ul style="list-style-type: none">• Designed and built high-performance, secure custom features and standalone PHP web tools for a large WHMCS-based hosting platform backed by MariaDB, working with cross-functional teams. | |
| Tech Lead Software Engineer
<i>SeoEra</i> | 06/2023 – 05/2024
Cairo, Egypt |
| Software Engineer
<i>SeoEra</i> | 12/2018 – 05/2023
Cairo, Egypt |
| <ul style="list-style-type: none">• Developed and maintained modern websites and admin panels using Laravel, Lumen, Vue.js, and Livewire, integrating authentication and authorization with Passport and Sanctum.• Designed dynamic, responsive front-end components with Vue.js, HTML, CSS, JavaScript, and jQuery, ensuring a smooth and engaging user experience across platforms.• Built robust back-end systems with MySQL and SQLite, utilized Laravel Nova for efficient admin management, implemented Socialite for social logins, and monitored applications using Laravel Telescope. | |
| Software Engineer
<i>@Designs Agency</i> | 01/2018 – 11/2018
Giza, Egypt |
| Java Developer & Web Developer
<i>Freelancer</i> | 01/2017 – 12/2017
Remote |

SELECTED PROJECTS

TrendX — Survey & Community Platform API

Laravel 11, PHP 8.2, MySQL, Redis, Elasticsearch, ClickHouse, Qdrant, MongoDB, Docker, REST APIs

Bilingual survey & rewards platform with a social community layer where organizations publish paid surveys and a community answers, posts, and earns rewards.

- Architected and built a layered Laravel 11 REST API (Controller → Service → Repository) with 190+ endpoints, strict typing, and PSR-12 enforced via Pint.
- Delivered the full survey lifecycle: creation, targeting, AI-assisted question generation and translation, response collection, analytics, and export.
- Integrated HyperPay for survey payments and YouGotAGift for a points-based reward and gift redemption system, including failure tracking and redemption finalization.
- Built a social layer with posts, polls, follows, topic feeds, boosts, impressions, blocking, and reporting, plus admin strike and moderation tooling.
- Implemented Elasticsearch-backed global search, Redis caching for high-traffic feeds, and a ClickHouse analytics pipeline streaming buffered answer, interaction, and impression events into denormalized fact tables.
- Shipped OTP and social authentication, role and permission based access control, ban/suspend enforcement, and spam and multi-device abuse detection.

YouMats – Multi-Vendor Building Materials Marketplace (v1 & v2)

Laravel 10/11, Nova, Octane, PostgreSQL, Redis, Docker, Vue.js

- Built and maintained a B2B/B2C e-commerce marketplace for construction materials serving the Saudi market, with multi-language, multi-currency storefront, city-based pricing, and SEO sitemaps.
- Integrated PayFort and Tamara (BNPL) payments, Shipa delivery, coupons, S3 media storage, OneSignal push notifications, and Pusher real-time events.
- Led the v2 rewrite as a headless REST API on Laravel 11 with Octane (Swoole), containerized with Docker, Nginx, PostgreSQL, and Redis.
- Designed a multi-region data layer with separate PostgreSQL databases per country (Saudi Arabia and UAE), and applied clean architecture using 40+ interface-bound repositories, enums, and service classes.
- Implemented three isolated Passport-authenticated domains (admin, vendor, user), cart/order/quotation flows with full status history, and Algolia search via Laravel Scout.

SKILLS

Programming Languages: PHP, Go (GoLang), JavaScript (ES2015+), Java (J2SE/J2EE), Dart, C++, SQL, Bash

Backend Frameworks: Laravel, Slim, Laravel Horizon, Laravel Passport, Laravel Socialite, OOP, MVC, SOLID, Design Patterns

APIs & Integration: RESTful APIs, GraphQL, gRPC, SOAP, WebSockets, OAuth 2.0, Webhooks, Payment Gateways, Third-Party API Integration

Databases (SQL & NoSQL): MySQL, PostgreSQL, SQLite, SQL Server, MongoDB, Redis, Elasticsearch (full-text search), ClickHouse, Qdrant (vector DB), Query Optimization, Database Design

Messaging & Queues: Apache Kafka, RabbitMQ, Laravel Queues, Event-Driven Architecture, Microservices

DevOps & Cloud: Docker, Docker Compose, Kubernetes, CircleCI, CI/CD, Nginx, Linux, SSH, Git, GitHub, DigitalOcean, AWS (EC2, S3)

Frontend & Mobile: Vue.js, jQuery, HTML5, CSS3, SASS, Bootstrap, Flutter, Ionic, Cordova

Testing & Practices: PHPUnit, Unit Testing, Integration Testing, API Testing, TDD, Code Review, Agile/Scrum, Clean Architecture

AI & LLM Integration: Claude Code, Codex, GLM, Cursor, Prompt Engineering (RAG, Few-shot, Chain of Thought), LLM-Powered Features, AI Content Moderation

CMS & E-commerce: WordPress, OpenCart, WHMCS

EDUCATION

Bachelor of Computer Science

MISR UNIVERSITY FOR SCIENCE AND TECHNOLOGY

2013 – 2017

Giza, Egypt